

Education

Tongji University, School of Chemical Science and Engineering

Sep. 2024 – Jun. 2028 (expected)

B.S. in Chemistry · National Top-Notch Student Training Program 2.0 Base for Basic Disciplines (Chemistry)

GPA: 4.53/5.00 (90.30/100) · **Rank:** 2/28 in the Top-Notch Program; top 5% in the School of Chemical Science and Engineering

Selected coursework: Inorganic Chemistry (Honors), Analytical Chemistry (Honors), Organic Chemistry, Physical Chemistry, Python Programming, and Foundations of Artificial Intelligence

Publication

Fluorenone Acceptor–Linker Engineering of Lead Halide MOFs for Aerobic Photocatalytic Oxidative Coupling

Dongyang Li, **Junkai Kang**, Fanyu Meng, Wenyan Dan*, and Honghan Fei*

Inorganic Chemistry (2026), accepted August 19, 2026 · DOI: [10.1021/acs.inorgchem.6c02797](https://doi.org/10.1021/acs.inorgchem.6c02797)

Research Experience

Conformational Design of Non-Fullerene Acceptors toward Low-Loss Organic Solar Cells

2026

Summer Research Student · Department of Chemistry, HKUST · Advisor: Prof. He Yan

- Investigated steric restriction of intramolecular rotation in non-fullerene acceptors as a molecular design strategy for modulating solid-state photoluminescence quantum yield (PLQY) and reducing non-radiative voltage losses in organic solar cells.
- Designed and synthesized parent and ortho-methyl-substituted tetraphenylethylene (TPE)-functionalized acceptors, using methyl-induced steric hindrance as the structural variable for investigating conformational control of solid-state emission.
- Executed literature-guided route planning, multistep synthesis, purification, and NMR characterization to establish the molecular basis for the comparative study of conformation and emissive properties.

Coordination-Directed Design of Hybrid Metal Halides for Photocatalysis

2025–2026

Undergraduate Researcher · Pb–Cu Project Lead · Tongji University · Advisor: Prof. Honghan Fei

- Synthesized a series of organic–inorganic coordination materials by varying ligands, metal sources, and halides, focusing on Pb- and Cu-containing systems to explore relationships between composition, crystal structure, and light-driven reactivity.
- Combined diffraction-based structural comparison and optical characterization across lead-halide coordination materials and Pb–Cu chalcogenides; photocatalytic testing identified activity within the series.
- Developed a two-stage Cu–thiolate/PbX₂ route for a representative Pb–Cu chalcogenide study targeting CO₂ photoreduction. Br-to-I substitution shifted the absorption edge from 546 to 557 nm and narrowed the optical gap from 2.41 to 2.35 eV.

ZIF-67 Nanozymes: Synthesis and Kinetic Analysis of Enzyme-Mimetic Activity

Sep. 2024 – Jan. 2025

Project Leader · Tongji University Innovative Experiment Design

- Prepared approximately 200 nm polyhedral ZIF-67 particles by room-temperature coordination self-assembly and characterized morphology and elemental distribution by SEM and elemental mapping.
- Quantified glucose oxidase-like and peroxidase-like activities by UV–vis kinetics and Michaelis–Menten/Lineweaver–Burk analysis; obtained an apparent Michaelis constant ($K_{m,app}$) of 0.05 mM for TMB.
- Received First Prize (1st place) in the Shanghai Undergraduate Innovative Experiment Design Competition.

Selected Honors and Awards

- **National Scholarship**, Ministry of Education of China (2025)
- **Third Prize**, National College Student Innovation Experience Competition (2024)
- **First Prize & Roadshow Champion**, Tongji University Innovation Experience Competition (2025)
- **Outstanding Student**, Tongji University (2025)
- **Third Prize**, “Guoyao Cup” Chemistry Competition (2024 and 2025)
- **Outstanding Camper**, Peking University Chemistry Camp (2022 and 2023; high school)
- **Gold Award**, China International College Students’ Innovation Competition, Tongji University (2026)
- **First Prize (1st place)**, Shanghai Undergraduate Innovative Experiment Design Competition (2025)
- **Tang Zhongying Scholarship**, Tongji University (2024 and 2025)
- **First Prize**, Tongji University Physics Competition (2025)
- **Third Prize**, China Undergraduate Mathematical Contest in Modeling (2024)
- **Third Prize**, Chinese Chemistry Olympiad (high school)

Technical Skills

Laboratory synthesis: Schlenk-line and inert-atmosphere synthesis; solvothermal synthesis; multistep organic synthesis; column chromatography.

Characterization and analysis: XRD/PXRD and profile comparison; UV–vis spectroscopy; NMR spectroscopy; SEM/elemental-mapping interpretation; enzyme-kinetics analysis.

Data and software: Python (pandas, NumPy, Matplotlib, SciPy); OriginPro; ChemDraw; Mercury; LaTeX; scientific data analysis and visualization.

Leadership and Service

- **Class President, Chemistry Top-Notch Program:** organized more than 30 class and university activities; two interdisciplinary education initiatives were selected as university key projects.
- **Peer Mentor, Qidi College:** supported junior students in academic planning, university transition, and early research exploration.
- **Student Union, Public Relations and External Liaison:** contributed to university-wide cultural events and intercollegiate coordination.
- **Service recognitions:** Outstanding Volunteer, Tongji University Student Union; Outstanding Class President and Outstanding League Cadre, School of Chemical Science and Engineering; Outstanding Individual, student Party branch initiative.
- **Community and science outreach:** completed approximately 200 hours across community education, environmental fieldwork, university outreach, and campus service.

Additional Experience

Technology Investment Intern, Bank of East Asia, Shanghai

2025

Academic Presentation and Science Communication

Delivered research presentations and chemistry-outreach sessions for high-school students in Henan (2024 and 2025).